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EC39002

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Experiment 4: Design of Clock Data Recovery (CDR) Circuits



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1 Objectives

The objectives of this experiment are to:

1. Design an elementary clock data recovery (CDR) circuit using a generated PN sequence as data.
2. Extract a synchronized clocking pattern prior to decision-making to generate the transmitted pattern synchronously.
3. Implement an edge detector and a stable Phase-Locked Loop (PLL) using the CD4046 IC to track frequency and suppress jitter.

2 Key Components

- **Integrated Circuits:**
  - IC CD4046 (Phase-Locked Loop)
  - IC 741 (Op-Amp)
- **Passive Components:** Resistors (100kΩ, 5kΩ, etc.), Capacitors (0.068μF, 47μF, 470pF).
- **Equipment:** Digital oscilloscope (Tektronix 2 Series), function generator, breadboard, connecting wires, and power supply.

3 Brief Theory

3.1 Clock Data Recovery (CDR)

In communication systems, clock and data recovery play a pivotal role, particularly in the operation of high-speed transceivers. These systems handle incoming data streams that are asynchronous and subject to noise, requiring the extraction of a synchronized clock for coherent operation.

Receivers employ a decision device to identify and estimate the transmitted bit pattern. The principal aim of the CDR circuit is to generate a repetitive clock signal with low distortion (i.e., low clock jitter) and to re-acquire lock in case of synchronization loss. The recovered clock must maintain a precise phase relationship with the data stream so that its rising edges align with the midpoint of each bit, ensuring maximum tolerance to jitter.

3.2 Phase-Locked Loop (CD4046)

The CD4046B is a Phase-Locked Loop (PLL) IC that contains a low-power, linear Voltage-Controlled Oscillator (VCO) and two phase comparators. The PLL tracks the phase of the incoming spikes from the edge detector. The loop filter (LPF) converts

phase error into a control voltage that adjusts the VCO. The loop gain must be carefully designed to ensure system stability and jitter suppression.

4 CDR System Architecture

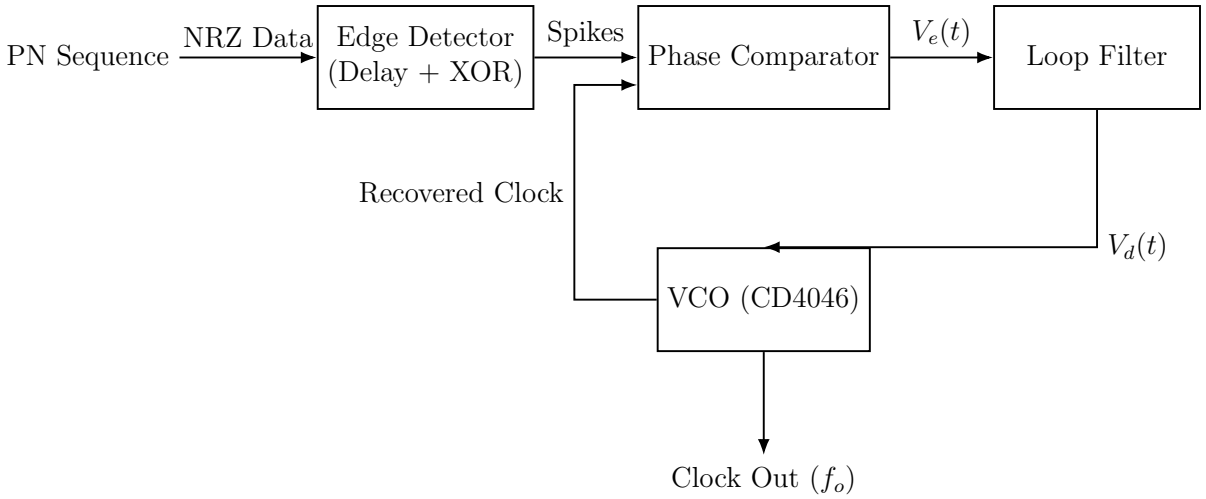


Figure 1: Block Diagram of the Clock Data Recovery Circuit using PLL.

The system consists of three main design problems:

1. **Edge Detector:** Extracts spikes from the input NRZ PN sequence using a delay block ($T_B/2$) and an XOR gate.
2. **Loop Filter:** Essential for the stability of the closed-loop feedback control system. For a PN-sequence clock of 10kHz, filter parameters are selected to ensure the VCO tracks frequencies from 0Hz to 70kHz.
3. **VCO Tracking:** The loop filter design must also suppress clock jitter without compromising stability.

5 Circuit Diagram

The complete hardware implementation of the CDR circuit is shown below, including the edge detector, loop filter, and CD4046-based PLL interconnections.

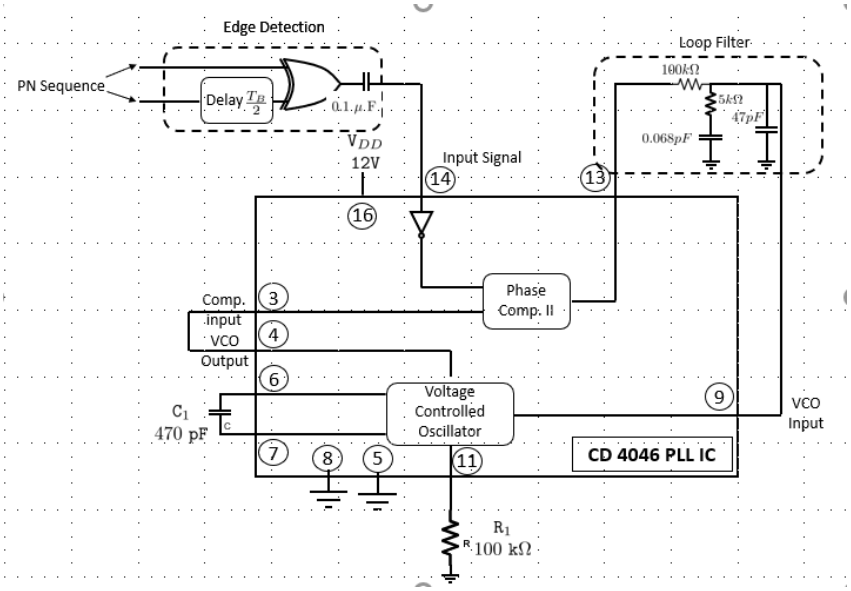


Figure 2: Complete circuit diagram of the implemented CDR system.

6 Observations and Results

6.1 Edge Detection

The edge detector converts the incoming Non-Return-to-Zero (NRZ) data into distinct impulses corresponding to the data transitions.

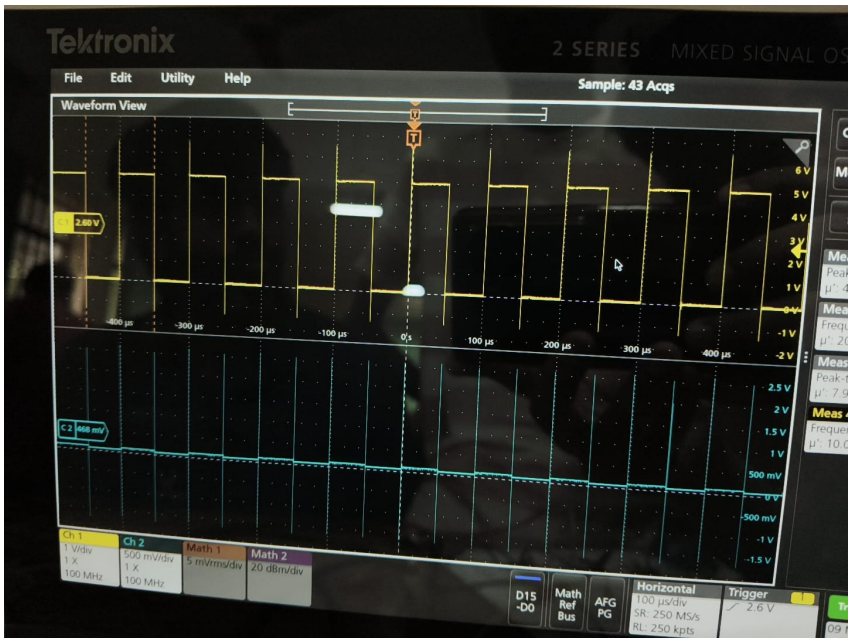


Figure 3: Oscilloscope Trace: Edge detection waveform. (Yellow: Input square wave, Blue: Edge impulse spikes generated at transitions).

6.2 Frequency Spectrum of Edge Impulses

To analyze the frequency components fed into the PLL, a Fast Fourier Transform (FFT) of the edge impulse signal was taken.

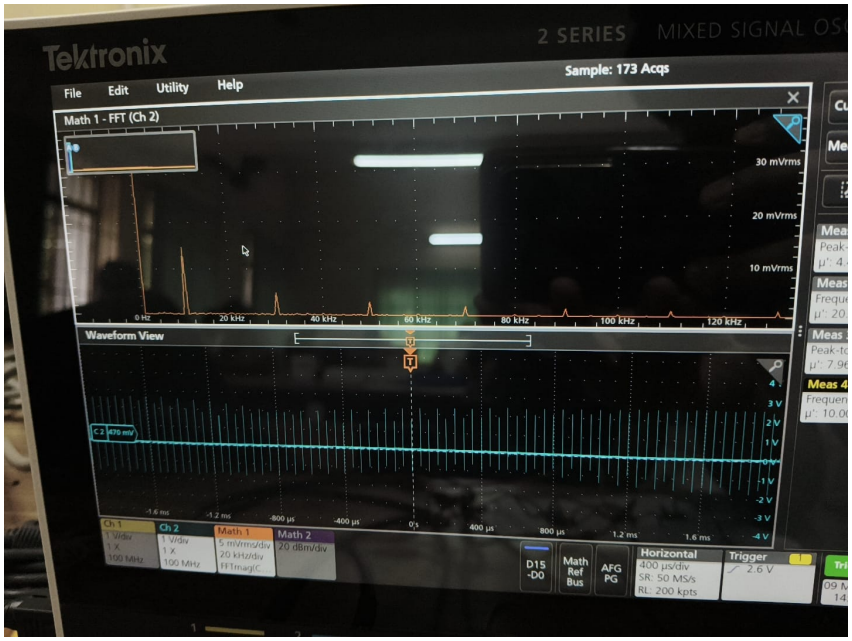


Figure 4: Oscilloscope Trace: FFT of the Edge Impulse showing the primary frequency components and harmonics.

6.3 Clock Recovery (PLL Output)

The Phase-Locked Loop successfully tracks the frequency of the edge impulses and outputs a continuous, synchronized clock signal.

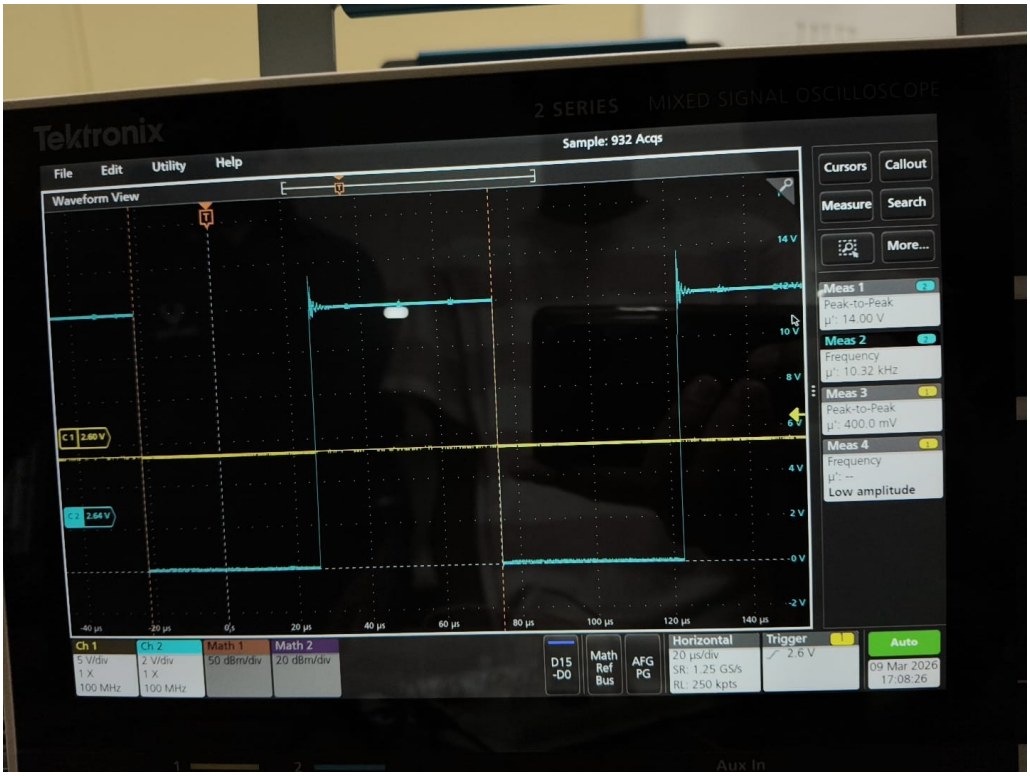


Figure 5: Oscilloscope Trace: Final Recovered Clock. (Yellow: Recovered clock, Blue: Input reference signal). The measured frequency stabilizes at approximately 10.32kHz.

7 Discussion

- **Challenges of NRZ and Edge Detection:** Recovering a clock sequence from a Non-Return-to-Zero (NRZ) signal is inherently more challenging than recovering one from a Return-to-Zero (RZ) signal. Unlike RZ, where the signal returns to a rest state within every bit duration, an NRZ signal remains constant during consecutive identical bits, providing no direct clocking information. To overcome this, we designed an edge detector to extract data-transition points. By passing the input PN sequence and a slightly delayed version of it (generated using cascaded NOT gates) through an XOR gate, we generated distinct phase pulses. Although these resulting impulse sequences are unevenly spaced due to the random nature of the data, this step is vital because the CD4046 phase comparator requires distinct spikes to lock.

- **Frequency Locking and FFT Analysis:** The uneven impulse sequence is fed into a Phase-Locked Loop (PLL), which locks onto the fundamental frequency of the data rate. As seen in the FFT of the edge-impulse signal (Figure 4), despite the signal's irregular time-domain appearance, there is a prominent fundamental frequency spike at 10 kHz, accompanied by diminishing harmonics. The PLL effectively isolates and locks onto this strong 10 kHz component, allowing the VCO to output a continuous, stable recovered clock.

- **Loop Filter Stability and Component Tuning:** The theoretical component choices for the passive low-pass filter provided in the experiment manual ($R_a = 5\text{k}\Omega$,

$R_b = 100\text{k}\Omega$) were initially tested but were found to be unstable in practice, failing to provide a steady control voltage $V_d(t)$ for the VCO. To achieve the stable lock shown in our final results, it was necessary to empirically tune these resistor values. This practical modification adjusted the feedback-loop gain and shifted the filter poles appropriately, ultimately stabilizing the control system and allowing the VCO to track the 10kHz input smoothly.

- **Clock Jitter Mitigation:** As seen in the recovered-clock waveform (Figure 5), the recovered clock aligns synchronously with the input with minimal observable phase variance. Proper tuning of the multi-pole loop filter helped mitigate high-frequency noise from the phase comparator, directly suppressing jitter in the final clock output.

- **System Assumptions & Limitations:** The primary assumption in this basic CDR circuit is that the data stream has sufficiently frequent transitions. Prolonged sequences of identical bits (long strings of '0's or '1's) produce no edges for the phase comparator, which eventually causes the VCO to drift from the target frequency. A realistic implementation requires line coding (e.g., Manchester encoding or 8b/10b) at the transmitter to guarantee high transition density.